Terafence
Reliable Transformations!

Complete Galvanic Network Isolation

for enhanced security and protection



TFG-121



Technology Overview

Designed to protect IoT and IIoT equipment, Terafence TFG-121 uses Air-Gap to segmentize and isolate industrial end devices from any harmful, or abusive, malware attacks and thus secure them from the network-based cyberattacks.

Terafence's proprietary hardware-architecture based on FPGA technology developed and manufactured by Terafene, creates a fully controlled data path between two network segments (OT / IT) and while allowing normal protocol data to flow from one side to the other, the return path simply does not exist, hence Air-Gapped.

Terafence is acting as a full protocol breaker / proxy server, terminating TCP/IP sessions on both ends and only moving raw data between the two unidirectional gateway sides. RAW data is not stored within the unit thus eliminating the requirement to safeguard such data by encryption or other methods. As nothing is stored, no such tools are used (like data encryption, compression or alike). Terafence technology and network mechanisms do not use cryptology to secure data exchange but instead denies network access.

Terafence TFG-121 not only protects IoT/IIoT end-devices from cyberattacks but also can protect other network assets by blocking any malicious attempts by an already infected external end-devices to infiltrate and infect devices with malicious code or cause other kind of damage.

Why choose TFG-121?

Key Features

- Proprietary Terafence FPGA-based hardware architecture for hardware-enforced security.
- Two dedicated CPUs (Atom E3940) for protocol support, processing, and termination.
- Secure HTTPS-based web GUI for configuration and management, accessible only from the secure side.
- Hardened operating system on the dedicated CPUs.
- Core security hardware operates independently without an operating system, MAC address, or IP address.

Solution Highlights

- Delivers physical network isolation through Terafence's proprietary FPGA-based hardware architecture.
- Supports protocol processing and termination through two dedicated proxy engines.
- Provides secure and intuitive configuration through a restricted HTTPS-based management interface.
- Designed for high availability, long-term operation, and a high MTBF (~20 years).
- Certified to IEC 62443 and MIL-STD-810F standards.

Supported Protocols

TFG-121 supports multiple, simultaneous industrial and IT protocols, including:
SFTP/FTP/SMB – Secure file or folder-tree transfers (10GB max)
SMTP – Email relay
SYSLOG Forwarding – Log event transmission
RTSP/ONVIF – Live streaming for CCTV (10 CCTV max)
HTTP/S – File and data uploads to web servers or cloud platforms
Local SYSLOG Forwarding – Internal log event management
Modbus – Communication between multiple PLCs and HMIs (one-way only)
IEC-104 – Protocol for power utilities (one-way only)
MQTT – Data brokering and publishing And various other protocols as per customer requirements.

Design:	Fanless architecture with no moving mechanical parts for enhanced reliability
Chassis:	IP30 protection, SECC, Black Metal
RAM / Local Storage:	8GB-DDR4/64GB-eMMC
Large-File Handling:	<10GB
Latency:	Protocol / data size dependent, core latency <2 nanoseconds

Technical Specifications

Product Type:	Unidirectional secure gateway / full protocol breaker with FPGA data diode – DIN rail / desktop version – fixed internal configuration
Data Throughput:	Up to 1 Gbps, auto-negotiation
Power Supply:	External 9–57 VDC @ 60 W, 2 inputs for redundancy
Network Interfaces:	2 × Ethernet (RJ-45) ports
Form Factor:	DIN rail
Mounting Options:	DIN rail
Dimensions:	82 × 200 × 160 mm
Operating Temperature:	32°F to 113°F
Storage Temperature:	-40°F to 185°F
Relative Humidity:	5 to 95%, 55°C (non-condensing)
Environment:	Indoor use only
Compliance:	IEC 62443, MIL-STD-810F, FCC, CE, EMI/EMC

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

info@terafence.us | www.terafence.us

© 2026 TERAFENCE USA Inc.